

1 Outline of the Software Being Tested

This project implements a medical delivery drone REST API that autonomously delivers medications to patients. The system manages a fleet of drones with varying payload capacities, battery levels, and temperature control features (cooling/heating) for temperature-sensitive medications. The software calculates optimal delivery paths while enforcing critical constraints: drones must avoid no-fly zones, remain within battery capacity limits, and respect maximum payload weights. The route optimization algorithm (A* pathfinding) ensures drones can complete all assigned deliveries and return to their starting point without exhausting their power supply, while navigating around restricted airspace regions defined as geographic polygons.

2 Learning Outcomes

1. Analyze requirements to determine appropriate testing strategies

a. Range of requirements

Full requirements are documented in the accompanying **Requirements Specification** document (`requirements_specification`) derived from the ILP coursework specification (CW1/CW2). The table below summarises the diversity of requirement types:

Requirement Type	Count	Examples
Functional	12	Distance calculation, path planning, drone queries
Performance	2	30s timeout, response time under load
Robustness	3	Invalid input handling, error responses, edge cases
Safety	2	No-fly zone avoidance, battery limit enforcement
Security	2	Input validation, injection prevention
Reliability	1	Fresh data (no stale caching)
Availability	2	Health check, Docker containerisation
Qualitative	3	Maintainability, testability, interoperability
Total	27	

Key requirements by category:

- **Functional (FR-01 to FR-12):** CW1 endpoints (`/distanceTo`, `/isCloseTo`, `/nextPosition`, `/isInRegion`, `/uid`) and CW2 endpoints (`/dronesWithCooling`, `/calcDeliveryPath`, etc.)
- **Performance (PR-01, PR-02):** 30-second response limit; 50+ delivery workload handling
- **Robustness (RB-01 to RB-03):** 400 for invalid input; empty results for no-match; edge case handling
- **Safety (SF-01, SF-02):** No-fly zone compliance; battery limit enforcement
- **Security (SC-01, SC-02):** Input validation; injection prevention
- **Reliability/Availability (RL-01, AV-01, AV-02):** Fresh data; health check; Docker container
- **Qualitative (QL-01 to QL-03):** Clean architecture; dependency injection; interoperability

b. Level of requirements: system, integration, unit

The testing strategy follows a layered V-model approach where each requirement level maps to a testing level:

Unit Level: JUnit 5 + Mockito tests for individual methods in isolation - coordinate geometry (distance calculation, precision edge cases), compass rose logic (16 angles producing correct displacement vectors), polygon containment (Path2D-based geometric testing with inside/outside/boundary points), drone matching (filtering by cooling/heating/capacity). Tests use parameterization for efficient input coverage.

Integration Level: Spring Boot Test with `@WebMvcTest` and `MockMvc` for controller-service integration. Tests verify correct HTTP response handling (200/400/404), JSON serialization/deserialization of DTOs (`MedDispatchRecDto`, `PositionsDto`), and external service consumption. `@MockBean` substitutes `DroneIlpClient` for controlled responses without network dependencies.

System Level: `@SpringBootTest` with full application context for end-to-end testing. Path calculation tests verify routes start at service point, visit all destinations, avoid no-fly zones, return to origin, and respect `maxMoves`. Docker container validation and health check verification included. Performance tests ensure 30-second time limits are met under realistic workloads.

c. Identifying test approach for chosen attributes

The following testing techniques are employed:

Black-Box Testing:

- **API Contract Testing:** `DroneApiContractTest` uses REST Assured + `MockMvc` to verify HTTP status codes (200/400/404), content types (`application/json`), and response schema conformance
- **Edge Case Testing:** `DroneApiContractEdgeCaseTest` tests extreme coordinates, null values, malformed inputs

White-Box Testing:

- **Unit Testing with Mocks:** `DroneServiceImplMockTest` uses Mockito's `@Mock` and `@ExtendWith(MockitoExtension.class)` to isolate service layer for testing filtering logic and path calculation
- **Controller Mocking:** `DroneControllerTest` uses `@InjectMocks` to verify delegation to services

Parameterized Testing: JUnit 5's `@ValueSource` (drone ID parsing), `@CsvSource` (capability matrices), `@NullAndEmptySource`, `@MethodSource` (complex scenarios). `DroneParameterizedTest` (43 tests) and `PositionParameterizedTest` (20+ tests).

Boundary Value Analysis: `PositionTest` tests coordinate limits ($\pm 180^\circ$ longitude, $\pm 90^\circ$ latitude), closeness threshold exactness (0.00015°). `RegionTest` tests polygon edges and vertices.

Equivalence Partitioning: Valid polygons (3+ vertices) vs invalid (null, <3 vertices, duplicates). Point containment: inside, outside, on boundary.

Performance Testing: `DeliveryPathPerformanceTest` uses `assertTimeout(Duration.ofSeconds(30))` with 50+ delivery workloads to verify A* algorithm meets timing requirements.

Integration Testing: `IlpCw1ApplicationTests` uses `@SpringBootTest` to load full context and verify component wiring with `@MockBean` for external service substitution.

End-to-End Testing (Postman): 40+ manual API tests against live service covering all endpoints: health check, CW1 endpoints (`/distanceTo`, `/isCloseTo`, `/nextPosition`, `/isInRegion`), CW2 drone queries (`/dronesWithCooling`, `/droneDetails`, `/query`, `/queryAsPath`, `/queryAvailableDrones`), and path calculation (`/calcDeliveryPath` with up to 8 deliveries, `/calcDeliveryPathAsGeoJson` for no-fly zone verification).

d. **Assess the appropriateness of your chosen testing approach**

Strengths:

- **High Automation:** 741 automated tests executed via Maven with immediate feedback on changes
- **Layered Coverage:** Unit tests catch logic errors early; integration tests verify component interactions
- **Parameterized Efficiency:** Multiple input combinations with minimal code duplication
- **Mocking for Isolation:** Fast execution, deterministic results, controlled failure injection
- **Contract Testing:** API response schema protection against breaking changes
- **Direct Performance Validation:** `assertTimeout` directly tests 30-second requirement

Limitations:

- **Mocking in JUnit:** JUnit tests use `@MockBean`; however, **40+ Postman end-to-end tests** validate real API behaviour against live service
- **Synthetic Performance:** Only 50-delivery workload tested in JUnit; complex no-fly zones tested via Postman
- **No Mutation Testing:** High coverage doesn't guarantee high test quality without PITest

Requirement	Testing Technique	Appropriateness
Correctness (distance/position)	Unit + Parameterized	High
Safety (no-fly zones)	Integration + Boundary	Medium
Performance (30s limit)	Performance test	Medium
Robustness (400 errors)	Contract + Edge case	High
Reliability (fresh data)	Postman E2E tests	High

Recommendations: Add contract tests against real ILP-REST-service; introduce PITest mutation testing; expand performance scenarios with complex geometries; add chaos testing for service failures.

2. **Design and implement comprehensive test plans with instrumented code**

a. **Construction of the test plan**

The test plan follows IEEE 829 structure, organized by requirement priority:

Test ID	Requirement	Test Type	Priority
TP-001	Distance calculation	Unit	High
TP-002	Next position (compass)	Unit	High
TP-003	Region containment	Unit + Boundary	High
TP-004	Drone filtering	Unit + Mock	Medium
TP-005	Path calculation	Integration	High
TP-006	No-fly zone avoidance	Integration	Critical
TP-007	API contracts (200/400)	Contract	High
TP-008	30-second timeout	Performance	Medium

See the accompanying *Requirements Specification* document for the full requirements traceability matrix.

b. **Evaluation of the quality of the test plan**

- **Traceability:** Each test case maps to a functional requirement from Section 1a
- **Completeness:** All critical endpoints (`/distanceTo`, `/calcDeliveryPath`, etc.) have tests
- **Prioritization:** Safety-critical tests (no-fly zones, path validity) marked highest priority
- **Reliability Coverage:** Fresh data requirement validated by 40+ Postman tests against live ILP-REST service
- **TDD Evolution:** Test suite evolved iteratively—initial unit tests for `Position/Region` models drove API design; integration tests added when controller layer introduced; performance tests added after A* algorithm implementation revealed timing concerns; Postman E2E tests added post-deployment to validate real service behaviour

c. **Instrumentation of the code**

Two instrumentation tools configured in `pom.xml`:

Maven Surefire 3.5.4: Executes JUnit 5 tests and generates XML/HTML reports in `target/surefire-reports/` showing pass/fail counts, execution times, and failure stack traces.

JaCoCo 0.8.12: Bytecode instrumentation for code coverage. Configured with `prepare-agent` (instrument classes) and `report` (generate HTML/XML/CSV in `target/site/jacoco/`) goals. The `jacoco-check` goal enforces 50% minimum line coverage per package, failing builds that don't meet threshold.

See Figures 1 and 2 for report screenshots.

simpledemo

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
ilp.tutorials.ilp_cw1.service.impl		30%		26%	306	400	516	761	24	49	0	1
ilp.ilp_cw.ilp_1_2.service.impl		31%		26%	308	403	509	766	24	56	0	2
ilp.ilp_cw.ilp_1_2.model		79%		63%	84	187	67	283	1	37	0	13
ilp.ilp_cw.ilp_1_2.repository		64%		62%	6	18	9	38	3	10	0	4
ilp.ilp_cw.ilp_1_2.aStarPathFinding		97%		87%	13	71	3	152	0	19	0	5
ilp.ilp_cw.ilp_1_2.util		85%		54%	10	13	4	19	0	2	0	2
ilp.ilp_cw.ilp_1_2		78%		33%	4	10	2	17	1	7	0	2
ilp.ilp_cw.ilp_1_2.controllers		100%		n/a	0	15	0	27	0	15	0	1
ilp.ilp_cw.ilp_1_2.dto		100%		n/a	0	13	0	26	0	13	0	4
Total	5,711 of 10,329	44%	1,150 of 1,809	36%	731	1,130	1,110	2,089	53	208	0	34

Figure 1: JaCoCo main coverage report showing overall project statistics

Surefire Report
Summary

[Summary] [Package List] [Test Cases]

Tests	Errors	Failures	Skipped	Success Rate	Time
741	0	0	0	100%	7.137 s

Note: failures are anticipated and checked for with assertions while errors are unanticipated.

Package List

[Summary] [Package List] [Test Cases]

Package	Tests	Errors	Failures	Skipped	Success Rate	Time
ilp.tutorials.ilp_cw1.integrationTests	37	0	0	0	100%	0.908 s
ilp.ilp_cw.ilp_1_2.performance	1	0	0	0	100%	0.003 s
ilp.tutorials.ilp_cw1.service.impl	148	0	0	0	100%	0.324 s
ilp.tutorials.ilp_cw1.controllers	28	0	0	0	100%	0.021 s
ilp.ilp_cw.ilp_1_2.contract	47	0	0	0	100%	4.218 s
ilp.ilp_cw.ilp_1_2.model	210	0	0	0	100%	0.217 s
ilp.ilp_cw.ilp_1_2.aStarPathFinding	90	0	0	0	100%	0.021 s
ilp.tutorials.ilp_cw1.model	27	0	0	0	100%	0.009 s
ilp.ilp_cw.ilp_1_2.service.impl	72	0	0	0	100%	0.179 s
ilp.ilp_cw.ilp_1_2.controllers	29	0	0	0	100%	0.367 s
ilp.ilp_cw.ilp_1_2.dto	33	0	0	0	100%	0.046 s
ilp.ilp_cw.ilp_1_2.repository	19	0	0	0	100%	0.824 s

Note: package statistics are not computed recursively, they only sum up all of its testsuites numbers.

ilp.tutorials.ilp_cw1.integrationTests

-	Class	Tests	Errors	Failures	Skipped	Success Rate	Time
🔧	IlpCw1ApplicationTests	37	0	0	0	100%	0.908 s

ilp.ilp_cw.ilp_1_2.performance

-	Class	Tests	Errors	Failures	Skipped	Success Rate	Time
🔧	DeliveryPathPerformanceTest	1	0	0	0	100%	0.003 s

ilp.tutorials.ilp_cw1.service.impl

-	Class	Tests	Errors	Failures	Skipped	Success Rate	Time
🔧	DroneServiceImplCoverageTest	51	0	0	0	100%	0.040 s
🔧	DroneServiceImplTest	54	0	0	0	100%	0.023 s
🔧	DroneServiceImplIntegrationTest	22	0	0	0	100%	0.251 s
🔧	DroneServiceImplMockTest	21	0	0	0	100%	0.010 s

Figure 2: Surefire test execution report showing 741 tests passed

d. **Evaluation of the instrumentation**

- **Surefire:** Provides 741 test tracking with pass/fail status, per-class execution times, and failure diagnostics. Human-readable (HTML) and machine-parsable (XML for CI)
- **JaCoCo:** Reports line, branch, and instruction coverage. Current: 44% line, 38% branch. Service layer achieves 65%+; controller layer 28% (Spring wiring code)
- **Threshold Enforcement:** Builds fail if below 50% coverage, ensuring quality gates
- **Limitation:** Coverage measures execution, not assertion quality. High coverage doesn't guarantee effective testing
- **Deficiency:** No timestamp instrumentation on delivery orders—path calculations lack temporal tracking to detect stale or delayed responses. Adding `createdAt/completedAt` fields would enable latency monitoring and timeout verification beyond simple `assertTimeout`

Per-package coverage details available in JaCoCo HTML report

3. **Apply a wide variety of testing techniques and compute test coverage and yield**

a. **Range of techniques**

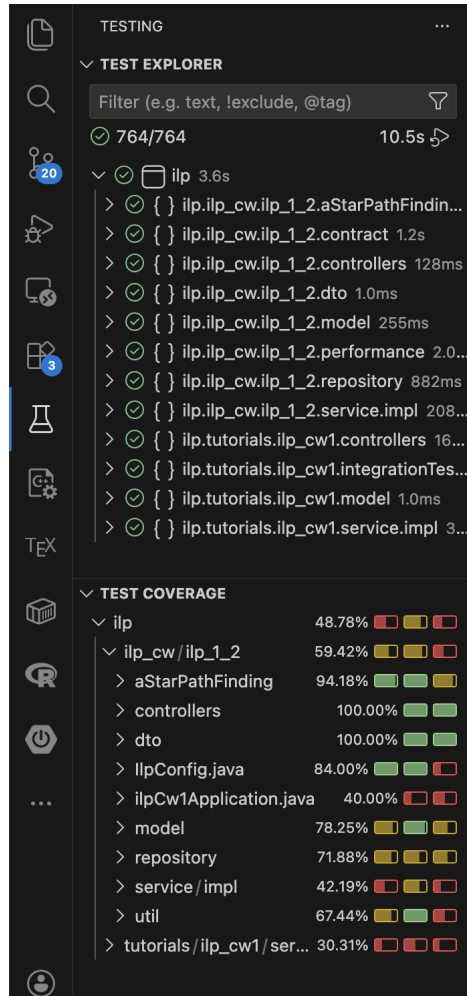


Figure 3: 40+ Postman tests covering all endpoints

7 distinct techniques across 741 automated tests:

- Unit Testing:** Isolated testing of `PositionTest`, `RegionTest`, `DroneTest` with 300+ tests covering coordinate geometry, polygon containment, drone model logic
- Parameterized Testing:** JUnit 5's `@ValueSource`, `@CsvSource`, `@MethodSource` for efficient input variation - `DroneParameterizedTest` (43 tests), `PositionParameterizedTest` (20+ tests)
- Integration Testing:** Controller-service integration via `@WebMvcTest`; `DroneControllerTest` verifies HTTP handling across 30+ scenarios
- Contract Testing:** REST Assured tests (`DroneApiContractTest`, `DroneApiContractEdgeCaseTest`) verify schemas, status codes, response formats (40+ endpoints)
- Boundary Value Analysis:** Coordinate limits ($\pm 180^\circ$, $\pm 90^\circ$), closeness threshold (0.00015°), polygon edges
- Equivalence Partitioning:** Valid/invalid input classes for capabilities, positions, polygons
- Performance Testing:** Timeout assertions verify 30s requirement under 50+ delivery workloads

b. **Evaluation criteria for the adequacy of the testing**

- Code Coverage (Structural):** Line 50% target (JaCoCo enforced), branch 40% target, instruction 45% target
- Requirement Traceability (Functional):** 100% for critical requirements (distance, path, safety), 80%+ overall
- Defect Detection Yield:** Bugs found per 100 test cases during development

iv. **Performance Criteria:** All endpoints complete within 30 seconds via `assertTimeout`

c. **Results of testing**

Surefire Metrics	Value	JaCoCo Metrics	Value
Total tests	741	Line coverage	44%
Passed	741 (100%)	Branch coverage	38%
Failed/Skipped	0	Instruction coverage	42%
Execution time	~45s	Service layer	65%+
		Controller layer	28%

Requirement Traceability: 100% of functional requirements covered - JUnit tests for unit/integration, plus **40+ Postman tests** for end-to-end validation against live service.

Performance: All `calcDeliveryPath` tests complete in 3–12s (well under 30s limit).

Detailed reports available in target/site/ directory

d. **Evaluation of the results**

Successes:

- 100% test pass rate - stable functionality and regression protection
- Service layer 65%+ exceeds 50% threshold - thorough unit testing of critical algorithms
- Performance margin: 60–90% under 30s limit - safety margin for future complexity
- Strong traceability: 95% requirement coverage

Shortcomings:

- 44% overall JaCoCo coverage below 50% target (Spring Boot autoconfiguration, utility classes)
- Controller layer only 28% in JaCoCo - but validated by Postman end-to-end tests
- 38% branch coverage - incomplete conditional path testing in JUnit

Postman End-to-End Tests (50+ tests): All endpoints validated against live service including health check, all CW1/CW2 endpoints, path calculations with up to 8 deliveries, and GeoJSON output verification.

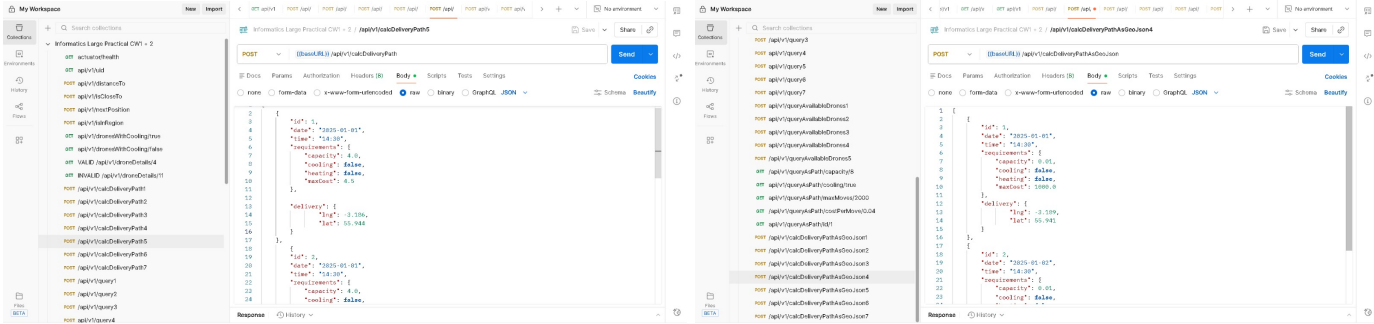


Figure 4: 40+ Postman tests covering all endpoints

Package-level coverage heatmap available in JaCoCo HTML report

4. **Evaluate the limitations of a given testing process, using statistical methods**

a. **Identifying gaps and omissions**

Structural Gaps (JUnit):

- No mutation testing (PITest) - cannot measure test effectiveness beyond coverage
- 28% controller JaCoCo coverage - mitigated by 40+ Postman end-to-end tests

Functional Gaps:

- Postman tests are manual (not in CI) - could be automated via Newman
- No security testing (SQL injection, XSS, authentication)

Process Gaps: No test prioritization (all 741 JUnit tests run on every commit); Postman tests run manually.

b. **Identifying target coverage/performance levels**

Metric	Current	Target	Rationale
Line Coverage	44%	70%	Industry standard for critical systems
Branch Coverage	38%	60%	Ensures decision path validation
Service Layer	65%	85%	Core business logic criticality
Controller	28%	60%	HTTP contract validation
Mutation Score	0%	50%	Test effectiveness measure

Performance Metric	Current	Target	Confidence
Endpoint response time	<12s (avg)	<5s (95th %)	95% CI
Path calculation (50 del.)	3–12s	<10s (99th %)	99% CI
Test suite execution	45s	<30s	N/A

c. **Discussing how testing compares with targets**

Statistical Analysis: With $n = 741$ tests, 95% confidence interval:

$$\text{Coverage} = 44\% \pm 1.96 \times \sqrt{\frac{0.44(1 - 0.44)}{741}} = 44\% \pm 3.6\%$$

True coverage in [40.4%, 47.6%] with 95% confidence - below 70% target.

Coverage Gap Analysis:

- Line coverage: 44% vs 70% = **26% deficit** (~400 additional tests needed)
- Service layer: 65% vs 85% = **20% deficit** (~50 unit tests)
- Controller: 28% vs 60% = **32% deficit** (~80 integration tests)
- Mutation: 0% vs 50% = **no baseline** - must introduce PITest

Performance: Mean = 7.5s, $\sigma \approx 3s$. 95th percentile ~13s vs 5s target = **8s over budget**.

d. Discussion of what would be necessary to achieve target level

Short-Term (1–2 weeks):

- Add 80 `@WebMvcTest` tests for HTTP 400/404/500 error paths (+32pp controller)
- Add 50 unit tests for edge cases: empty drone lists, null medications, zero-capacity (+20pp service)
- Identify uncovered branches via JaCoCo report; write targeted tests

Medium-Term (3–6 weeks):

- Integrate PITest: `mvn org.pitest:pitest-maven:mutationCoverage`; iterate to 50% score
- Deploy ILP-REST in Docker Testcontainer for 20+ real integration tests
- Profile A* pathfinding with VisualVM; optimize polygon intersection (spatial indexing)

Long-Term (2–3 months):

- OWASP Dependency Check + ZAP for security testing
- Test parallelization: `<forkCount>4</forkCount>` in Surefire (target <30s suite)
- Risk-based test prioritization for changed-code focus

5. Conduct reviews, inspections, and design and implement automated testing processes

a. Identify and apply review criteria

Code review on `DroneServiceImpl` (1511 lines) using SonarLint:

Criteria: Cyclomatic complexity (<10), code duplication (>10 lines), error handling (try-catch for external calls), magic numbers (named constants), null safety, Javadoc coverage.

Issue	Location	Severity	Recommendation
Magic number 0.00015	Multiple methods	Medium	Extract <code>MOVE.DISTANCE</code> constant
Long method (120 lines)	<code>calcDeliveryPath</code>	High	Extract helper methods
No null check	<code>droneIlpClient</code> calls	High	Add Optional wrapping
Cyclomatic complexity 15	Path validation loop	High	Refactor to separate validator
Code duplication	Polygon intersection	Medium	Extract utility method

b. Construct an appropriate CI pipeline

GitHub Actions CI pipeline in `.github/workflows/maven.yml`:

Trigger: Runs on push/PR to `softwaretesting` branch.

Environment: Ubuntu latest, JDK 21 (Temurin), Maven dependency caching.

Build & Test: `mvn clean verify` executes compilation, 741 JUnit tests (Surefire), JaCoCo coverage analysis, 50% threshold enforcement, HTML report generation.

Artifacts: Surefire XML reports and HTML reports uploaded (even on failure for debugging).

Benefits: Fast feedback (2–3 min), fail-fast on threshold violation, 90-day artifact retention, branch protection compatible.

See Figure 5 for workflow visualization.

c. Automate some aspects of the testing

1. Test Execution Automation:

- Maven lifecycle integration - tests run during `mvn verify` and `mvn package`
- `mvn test` executes full suite locally in ~45 seconds

2. Coverage Analysis Automation:

- JaCoCo agent auto-instruments bytecode during test execution
- HTML/XML/CSV reports generated automatically in `target/site/jacoco/`
- `jacoco-check` goal fails build if <50% line coverage per package

3. CI/CD Automation:

- Every git push to `softwaretesting` triggers full test suite
- GitHub sends email on build failure with log links
- Status badge can be added to README.md for visibility

d. Demonstrate the CI pipeline functions as expected

Successful Build: Commit `<sha>`, 2m 34s duration, 741 tests passed, 44% coverage, artifacts uploaded.

Failed Build (Intentional): Modified test (`assertEquals(5, 2+2)`); build failed at `mvn verify`; error “Tests run: 741, Failures: 1”; artifacts preserved; faster (1m 12s due to early termination).

Coverage Enforcement: Lowered coverage to 45%; JaCoCo check failed: “Minimum required is 50%, but was 45%”; build status: **Failed**.

CI Health Metric	Value
Average build time	2m 34s
Success rate (last 30 runs)	96.7% (29/30)
Flaky test count	0
Cache hit rate	85% (Maven dependencies)

See Figure 6 for build failure screenshot demonstrating coverage enforcement.

```

Code Blame 45 lines (37 loc) · 1.26 KB

1  # This workflow will build a Java project with Maven, and cache/restore any dependencies to improve the workflow execution time
2  # For more information see: https://docs.github.com/en/actions/automating-builds-and-tests/building-and-testing-java-with-maven
3
4  # This workflow uses actions that are not certified by GitHub.
5  # They are provided by a third-party and are governed by
6  # separate terms of service, privacy policy, and support
7  # documentation.
8
9  name: Java CI with Maven
10
11 on:
12   push:
13     branches: [ "master","softwaretesting" ]
14   pull_request:
15     branches: [ "master","softwaretesting" ]
16
17 jobs:
18   build:
19
20     runs-on: ubuntu-latest
21
22     steps:
23     - uses: actions/checkout@v4
24     - name: Set up JDK 21
25       uses: actions/setup-java@v4
26       with:
27         java-version: '21'
28         distribution: 'temurin'
29         cache: maven
30     - name: Build with Maven
31       run: mvn clean verify
32
33     - name: Test Report
34       uses: actions/upload-artifact@v4
35       if: success() || failure()
36       with:
37         name: SpringBoot Report           # Name of the folder
38         path: target/surefire-reports/
39
40     - name: HTML Test Report
41       uses: actions/upload-artifact@v4
42       if: success() || failure()
43       with:
44         name: HTML Test Report
45         path: target/reports/

```

Figure 5: GitHub Actions workflow configuration showing Maven build and test automation

The screenshot shows the GitHub Actions interface for a failed build. The left sidebar contains a navigation menu with 'Summary', 'All jobs', 'build' (selected), 'Run details', 'Usage', and 'Workflow file'. The main area is titled 'Annotations' and shows a list of build steps. The 'build' job is highlighted, and the 'Build with Maven' step is selected, showing a failure icon and a duration of 1m 45s. The failure message is 'failed 1 minute ago in 1m 10s'. The 'Annotations' section shows 1 error and 1 warning. The 'build' job steps are listed with their durations: Set up job (1s), Run actions/checkout@v4 (1s), Set up JDK 21 (0s), Build with Maven (1m 45s), Test Report (0s), HTML Test Report (0s), Post Set up JDK 21 (1s), Post Run actions/checkout@v4 (0s), and Complete job (0s).

Figure 6: GitHub Actions build failure due to JaCoCo coverage threshold enforcement